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Stress Detection using Wearable Physiological Sensors and Machine Learning Algorithm

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***Abstract*: The primary objective of this methodology is to develop and test the performance of a wearable psychological sensor using EEG, EDA and ECG to observe the stress level of humans and analyze if the observed variation in psychological signals is related to the objective biomarker of stress. An integrated wearable system with physiological sensors is designed, developed, tested and evaluated in this paper for monitoring stress using biological markers in the working environment. Stress detection is performed using the Muse S (Gen 2) EEG headset, Savvy ECG sensor, and Shimmer3 GSR sensors. For each subject under test, 32 features are extracted from the multi-modal signals, of which, five features are retained for EEG, four signals for ECG and 1 for EDA, summing up to 10 features. The window size for the collection of each feature is one minute. Hence, for the 20 participants and ten features, a data of 200 minutes is used as training data in a controlled environment. Of these, 138 minutes are labelled as stressful tasks and 62 minutes are labeled as no-stress. The proposed stress detection model effectively assesses stress under a controlled environment with an accuracy of 97% and in an everyday environment with an accuracy of 93%. This work aims to design and develop a remote control system that can be used in medical devices to prevent the aftereffects of stress.**

Keywords— Stress Management, Therapy, EEG, ECG, EDA, Stress

I. Introduction

With an increase in the population and the subsequent decrease in healthcare workers, there is a high need to balance this ratio. The Organization for Economic Co-operation and Development (OECD) has warned the human race about the possibility of a shortage in the number of doctors and health workers in the near future [1]. This has led to the invention of several novel technologies that aim to monitor people's mental and physical health, supporting doctors, caregivers and health workers. This technology is called Quality of Life Technologies (QoLTs) [2]. This technology involves applying findings from various technological areas to improve people's quality of life. Self-therapy and psychology are two of the most emerging research topics in QoLTs, which aim to improve the quality of life. Many technologies already exist to provide psychological support to people. However, the mental health level is here to be treated appropriately, and technologies for the same are very limited [3]. In recent years, treating people's adverse mental health, specifically stress, has gained much attention, and there is a need for developing technologies to address this core issue.

Stress has become an inevitable part of our everyday life, and there is an increase in the number of stressful situations a person faces during their daily activities. For an outside perturbing factor, stress is a natural reaction by our human body [4]. Electro-dermal activity, pupil dilation, skin temperature, blood volume pulse and heart rate are some physiological responses to stress. Though stress does have beneficial characteristics that improve our reflexes, research shows that long-term stress is related to several health issues, such as premature aging and depression. A survey in India by LinkedIn released during the month of December 2020 indicated that 74% of Indians suffer from stress, wherein 2% reported severe stress, 4% faced moderately severe symptoms of stress, 11% were moderately stressed, and 57 per cent respondents were suffering from mild stress [5].

Similarly, a survey in the UK indicates that about 12 million people in the UK pay a regular visit to the general practitioner doctor to examine and express their mental health issues. This, in turn leads to a loss of 13.3 million working days. Considering these factors, a stress management technology needs to be invented. In recent years new wearable technologies have been invented to monitor and alert people on the amount of stress they undergo in their everyday life. Hence this work indicates the need and requirement for the design and development of novel wearable technologies [2] which monitor people and determine their stress levels with respect to their everyday activities. In this work, an integrated wearable system with physiological sensors is designed, developed, tested and evaluated in this paper for monitoring stress using biological markers in the working environment. Stress detection is performed using the Muse S (Gen 2) EEG headset, Savvy ECG sensor, and Shimmer3 GSR sensors.

II. Literature Review

Electrodermal Activity (EDA) is also known as Skin Conductance (SC) and Galvanic Skin Response (GSR) and is the biomarker of sympathetic nervous system activation. Skin Conductance is considered a highly valid marker that depicts sensitive information regarding emotional arousal [3]. The GSR sensor is used to determine the level of emotional arousal based on the secretion of sweat detected on the feet and hands of the person. Skin conductance up to the point of saturation and arousal are directly proportional to each other. In general, there are two types of skin conductance, namely skin conductance response (SCR) [4] and skin conductance level (SCL) [5]. SCR, or phasic skin conductance, refers to the body's reaction when a stimulus is given. On the other hand, tonic skin conductance or SCL is the observation of baseline level in an experiment without environmental events. The stress level can be analyzed using several GSR features, such as

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- Rising time which is the time difference between the peak of the response and the onset of the stimulus.
- Latency which is the temporal distance between the start of the response and the stimulus.
- The amplitude of the response is the difference between SCL at pre-stimulus and SCR at the peak of the response.

In general, the temperature of the skin varies between 32 and 35-degree Celsius. In case of physiological changes, physical exertion, malnutrition, fever and extreme environmental temperature, higher variation in the skin temperature [5]. The skin's temperature can be reliably and efficiently measured with a temperature sensor by placing it in contact with the skin. This has enabled skin temperature monitoring and has been studied by researchers for over a decade. However, there is much debate over the prediction of stress based on skin temperature. Some studies indicate that the skin temperature increases in the presence of stress. On the other hand, other studies indicate that in the presence of stress the temperature of the skin decreases [4].

Similarly, stress measurement is done with the help of heart rate variability (HRV). The HRV indicates parasympathetic and sympathetic measurements based on the temporal difference between subsequent heartbeats. Inter-beat interval (IBI) and R-R interval represent the time between beats, measured in milliseconds. Root-mean square, standard deviation RR intervals, mean value etc. The most commonly used methodology [6] is the LF/HF ratio, high frequency (HF) and low frequency (LF). A survey on the various press diction strategies used to analyze the psychological signals for emotional stress indicates an array of approaches. Several studies show methodology wherein the patient under test is first identified to be stressed or not. In this case, the patient is requested to solve arithmetic problems within a short period in the presence of a social evaluative threat for psychological stress. GSR features were fed to SVM, which resulted in an accuracy of 82.8% [7].

The values of ST, GSR and SVP, were measured by Zhai and Barreto in [8] using the stressor Stroop Test. Based on their investigation, SVM was identified as the most accurate learning algorithm, with an attainment of 90.1% accuracy. Similarly, in [9], the authors introduced an emotional stress recognition system with the help of respiration rate, BVP, EEG and GST. Based on their analysis, an accuracy of 82.7% was attained for SVM and elicit stress. Authors in [16] presented work for stress identification using respiration rate, EMG. GSR, HRV, and Heart rate (HR) in several situations with stressors in varying sound levels. Thus, it is concluded that several physiological signals can be used in combination or alone to identify stress. Despite these observations, it is to be noted that an increase in the count of physiological signals is not a means of ensuring the highest accuracy. For example, four different signals were used by the authors Wijsman et al. in [10], resulting in an accuracy of 74.5% in contradiction to the 82.8% accuracy attained by Setz et al. in [11], who used only GSR values.

Moreover, the chosen algorithm also plays a significant part in the combination of physiological signals. It has been identified that the GSR value is a primary value used by all researchers in combination with BVP or ECG. Table 1 shows the domination of SVM, and it is observed that a combination of ANFIS, GRNN and MLP is used to attain the highest accuracy. EMG and GSR were the right matches for identifying and evaluating stress levels. Higher accuracy in stress level classification is identified with fuzzy logic and feature-based algorithms [12]. Though an accuracy of 95% is attained by several researchers, for emotional states classification, the accuracy attained is curbed at 90%. A summary of the researchers, the physiological signals and the accuracy level are depicted in table 1.

Table 1. Stress Detection Survey

Reference	Physiological Signals	Method	Stressor	Accuracy
Zhang (2018) [13]	EMG, ECG, GSR	SVM	Auditory stimuli and Stroop test	88.5%
Cho et al. (2017) [14]	GSR	Machine Algorithm with Kernel	Arithmetic subtractions in configurable Virtual Reality	95%
Keshan et. al (2015) [15]	EEG	Random Tree	-	88.2%
Jun and Smitha (2016) [16]	EEG	SVM	Mental arithmetic test and Stroop test	75%
Wijsman et al. (2013) [17]	RESP, EMG, ECG, and GSR	General Estimating Equations	Auditory	74.5%
Sharma and Gedeon (2013) [18]	BVP, ECG and GSR	ANN with SVM	Exposure to texts	89%
de Santos Sierra et al. (2011) [19]	ECG	Fuzzy logic	Talk preparation and Hyperventilation	99.5%
Kolodyazhniy et al. (2011) [20]	ST, EMG, ECG and GSR	KNN	Films	84.5%

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III. Proposed Work

An integrated wearable system with physiological sensors is designed, developed, tested and evaluated in this paper for monitoring stress using biological markers in the working environment. Stress detection is performed using the Muse S (Gen 2) EEG headset, Savvy ECG sensor, and Shimmer3 GSR sensors. The EEG, ECG and EDA features are evaluated based on their statistical significance to improve the reliability and overall accuracy of stress detection. Physical stress is induced through acute stress manipulation (Maastricht Acute Stress Task, Montreal Imaging Stress Test and the Scan STRESS techniques. These are clinically certified and well-defined stress induction programs. The data from the wearable devices are further processed to identify if the user is stressed or not. If the user is stressed, the level of stress is identified.

A. Experimental setup

A multimodal sensor infrastructure is essential to detect and analyze the variations in physiological signals. A dedicated interface is used for connecting a personal computer and three different wearable sensors, namely the Muse S (Gen 2) EEG headset, Savvy ECG sensor, and Shimmer3 GSR sensors used in this setup. These sensors used for analysis are shown in figure 1. The Muse S (Gen 2) EEG headset uses Bluetooth 4.2 and weighs 41 grams. The Savvy ECG sensor uses two electrodes placed on the chest and weighs 21 grams. It enables continuous measurement of ECG signals and continuous data transfer with total freedom to move for the user due to its wireless nature. The Shimmer3 GSR sensor uses the EDA to provide the Electrodermal Resistance (EDR) measurement, also called the Galvanic Skin Response (GSR). It consists of an optical pulse sensing probe that can be attached to the finger, optical pulse sensors that can be connected to the earlobe, GSR and two electrodes, two biophysical 9″ leads (x2) and one wrist strap. Preamplification is performed while acquiring this data. The Photoplethysmogram (PPG) and the optical pulse signals are captured based on the electrical conductance and characteristics of the skin.

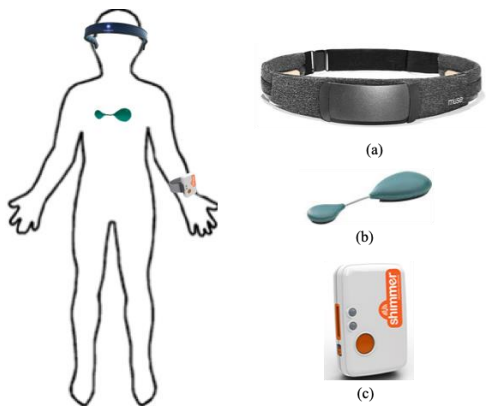

Fig. 1: Left: Wearable sensors used to acquire physiological signals and the way to be worn; Right: (a) Muse S (Gen 2) EEG headset, (b) Savvy ECG sensor, (c) Shimmer3 GSR Sensor.

a) Physiological Data Collection

The wireless sensor systems acquire the physiological data during the experimental tests. The EEG, EDA and HRV signals are measured using wearable sensor devices. The Muse S (Gen 2) EEG headset captures the EEG brain waves, breathing and heart rate to recognize and visualize fatigue, stress, sleep quality and concentration. Along with EEG, it also supports the biofeedback through Photoplethysmogram (PPG), Accelerometer, Gyroscope, EFG and smart-fade technology. Physical and mental stress causes variations in brain activity. Different physical and mental stress states are represented by several frequency bands in which the EEG signals are arranged. The Savvy ECG sensor enables continuous measurement of ECG signals. During each cardiac cycle, the heart's electrical activity is measured with the help of ECG signals. The depolarization and repolarization of the heart produce electrical signals captured by the ECG electrodes. A T wave, QRS complex and P wave are typically present in the ECG signals. The distance between adjacent R peaks is called R-R interval, which is used to estimate the heart rate variability (HRV) and heart rate (HR).

The EDA measures the electrical conductivity of the skin, called the GSR. The unobtrusiveness and measurement accuracy of the sensors is analyzed. The brain, electrodermal and cardiac activities are unobtrusively measured by these devices, such that a minimum level of acceptability is guaranteed from the user's perspective. The galvanic resistance is obtained from the Shimmer sensor, converted into galvanic SC. The phasic activity varies in a relatively faster manner compared to the tonic activity in the SC. The local peaks of the phasic activity startle for a duration of 1 to 5 seconds, which is evaluated to obtain the GSR. The noise from this signal can be removed using a low-pass Butterworth filter with a cut-off frequency of 1Hz. The root mean square, area, duration, amplitude, rise time and fall time of the signals are estimated using a trough-to-peak method.

b) Participant Screening

A random population of 20 healthy participants from Coimbatore, Tamil Nadu, India, were enrolled for data collection, of which there were ten males and ten females with a mean age of 35.2 ± 15.3 years. People exposed to medication that may affect the stress response, substance abuse, heavy smoking, diabetes, hypertension, chronic heart failure, coronary artery disease and cardiac arrhythmias are excluded based on an assessment by a physician. The participants are exposed to physical stress induced through an acute stress manipulation technique such as Maastricht Acute Stress Task, Montreal Imaging Stress Test and ScanSTRESS.

B. Experimental setup

The data must be preprocessed before feature extraction. Filtering and smoothing are the two steps in preprocessing the noise removal signals. The noise removal technique is performed as shown in figure 2. The EEG, EDA and HRV signals are processed with the help of the appropriate digital signal processing technique. The EEG signals are low

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clarity as the electrodes are placed on the scalp. The signals are denoised using a Butterworth bandpass filter and enhanced through data segmentation. 20 EEG recordings are collected, of which 20 segments are retained for each signal, with each segment of length one second per subject. The signals are further classified based on their frequencies into alpha, beta, theta, gamma and delta. The NumPy library is used to plot the FFT graph of the time signals to extract the features. The data is analyzed in both the time and frequency domain using the matplotlib function specgram. The spectrogram helps visualizing the EEG signal in time, frequency and power domains.

The noise in the ECG data is also removed using a bandpass filter. Further, a moving average filter is used to smooth the data. The mean value of HR is extracted from the filtered ECG data. The non-linear HRV variables and the mean value of the time domain are also extracted. The Power Spectral Density analysis and Fast Fourier Transform are used for analyzing the power distribution as a function of frequency in the frequency-domain HRV variables. This enables the quantification of an atomic balance at any specific time. The mean of the frequency-domain HRV variables is extracted after preprocessing. The PPG data is processed using the moving average and bandpass filters for noise removal and smoothing of the data. The mean HR value is estimated from the filtered PPG data. The non-linear, frequency and time domain HRV variables are extracted using the same feature extraction functions. Median and moving average filters are used for processing the GSR signal for extracting the skin conductance. Noise removal is performed using the median filter, and the data is smoothened using the moving average filter. The mean value of skin conductance is extracted after preprocessing. Variation is calculated by extracting the skin conductance gradient.

A smartphone app is used for collecting wearable sensor data via Bluetooth. The emotions and stress levels are perceived subjectively using the data collected by the app. The app collects the physiological signals to form a database, and timestamps are added to all the measurements based on the time on the smartphone. The global navigation satellite system (GNSS) sensor of the phone helps geolocate the data for each participant. The real-world experiments on a spatial context and the associated stress response data are analyzed through this.

C. Data Preprocessing and Feature Extraction

The data must be preprocessed before feature extraction. Filtering and smoothing are the two steps in preprocessing the noise removal signals. The noise removal technique is performed as shown in figure 2. The EEG, EDA and HRV signals are processed with the help of the appropriate digital signal processing technique. The EEG signals are low clarity as the electrodes are placed on the scalp. The signals are denoised using a Butterworth bandpass filter and enhanced through data segmentation. 20 EEG recordings are collected, of which 20 segments are retained for each signal, with each segment of length one second per subject. The signals are further classified based on their frequencies into alpha, beta, theta, gamma and delta. The NumPy library is used to plot the FFT graph of the time signals to extract the features. The data is analyzed in both the time and frequency domain using the matplotlib function specgram. The spectrogram helps visualizing the EEG signal in time, frequency and power domains.

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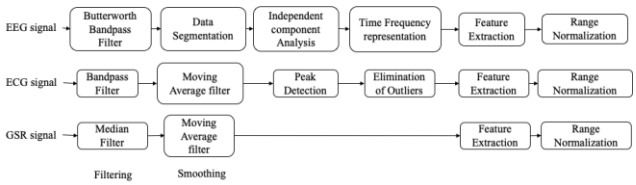

Fig. 2: Preprocessing and Feature Extraction process

D. Feature Selection

Using all features from the extracted signals is unnecessary as it may not help improve the accuracy and will cost a time overhead during data processing. The noise component may also increase when a feature unrelated to stress is along with the related features. The computational complexity may increase when loosely correlated features are computed. For example, non-linear computational complexity is observed in the frequency domain and independent features. Hence, the features correlated to stress are only selected in this stage. A greedy stepwise algorithm is adopted to identify the best subset of features. This technique starts with an empty set and adds features that enhance the accuracy while removing features that decrease it. These steps are repeated until a set of features is obtained such that adding or deleting any selected features does not improve the accuracy further. Information gain, correlation-based feature selection technique and k-nearest-neighbor classifier are used to estimate each subset's accuracy. The features selected based on this technique are shown in table 2.

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TABLE 2 FEATURES EXTRACTED FROM THE SENSORS WITH SELECTED FEATURES IN BOLD

Sensor	Features
EEG	Mean, median, variance, Standard deviation, Skewness, Kurtosis, Zero-crossing rate, number of waves, wave duration, peak amplitude, Instantaneous frequency, mobility, activity, complexity, alpha, beta, theta, gamma, delta
ECG	LF, HF, LF/HF, SD1, SD2, SD1/SD2, pNN20, pNN50, HR, SDRR, SDSD, RMSSD
EDA	Skin conductance

Here, HF represents the high-frequency band ranging between 0.04 and 0.15 Hz, LF represents the low-frequency band ranging between 0.15 and 0.4 Hz, the LF to HF ratio is represented by LF/HF, SD1 and SD2 represent the standard deviation perpendicular, and along the line of identity respectively, the SD1 to SD2 ratio is represented by SD1/SD2, pNN20 and pNN50 represents the percentage of adjacent intervals differing from each other by over 20ms and 50ms respectively, the standard deviation of R-R intervals is represented by SDRR, between adjacent R-R intervals the standard deviation and root mean square of successive differences is given by SDSD and RMSSD respectively.

E. Classification using Machine Learning

When general data sets and personal datasets are used, the bias of physiological data can show substantial variations. Personal dataset with the data gathered from the 20 participants is used for testing the efficiency in classifying the extracted features. Naive Bayes classifier, support vector machine (SVM), K-nearest-neighbor (kNN), Artificial Neural Networks (ANN) and Random Forest Classifier (RFC) are the machine learning algorithms used for the classification of the signals. A comparison of these algorithms in terms of classification accuracy is estimated.

IV. RESULTS AND DISCUSSIONS

The experimental results are observed in a controlled setting as well as in everyday settings. NB, SVM, kNN, ANN and RF classifiers validate the stress models. A comparison of these models is shown in figure 3. The data points are tested across all subjects to check if they are generalized by the classifiers. Everyday data is used for applying the classifier and predicting stress. The contextual factors that affect the results are observed and studied. The results are analyzed, and mitigation techniques can be proposed. The reference point is set based on the stress labels obtained by the participants on an everyday self-report. Context data while performing activities such as eating, walking, running and so on are also collected, and the effect of motion artefacts and such noise on the decisions in the everyday environment is evaluated. The wireless sensor systems acquire the physiological data is grouped into six categories, namely, EEG, ECG, EDA, EEG + ECG, ECG + EDA, and EEG + ECG + EDA. They are analyzed to see how stress detection accuracy varies for a combination of features. EDA is cost-effective and can be worn conveniently in everyday settings, as shown in figure 4.

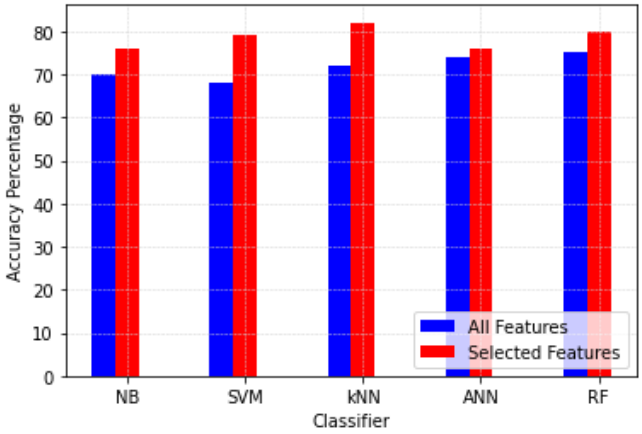

Figure. 3: Comparison of cross-validation accuracy of various classifiers using all features and features

Testing and training data are categorized. Classification and prediction are performed using Spyder IDE and Python language. For each subject under test, 32 features are extracted from the multi-modal signals, as shown in table 1. Of these, five features are retained for EEG, four signals for ECG and 1 for EDA, summing up to 10 features. The window size for the collection of each feature is one minute. Hence, for the 20 participants and ten features, a data of 200 minutes is utilized as training data in a controlled environment. 138 minutes are labelled as stressful tasks, and 62 minutes are labelled as no stress. The proposed stress detection model effectively assesses stress under a controlled environment with an accuracy of 97% and in an everyday environment with an accuracy of 93%. Binary values are used for labelling if the participant is stressed or not. With advanced machine learning techniques, stress assessment with finer granularity can be attained in the future where high, medium, low, and no stress labels that are fine-grained can be used for labeling the outputs of the stress tests.

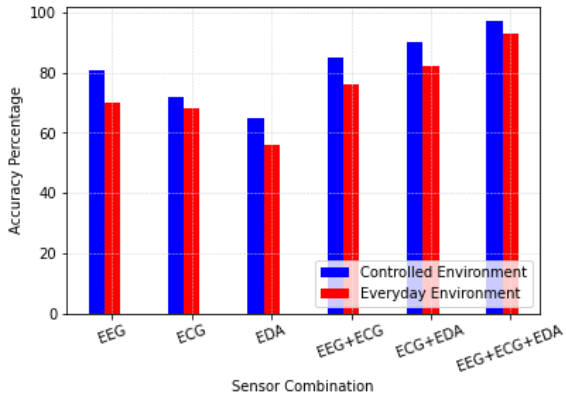

Fig. 4: Performance comparison of the combination of features in controlled and everyday environment

V. CONCLUSION AND FUTURE SCOPE

The performance of wearable psychological sensors using EEG, EDA and ECG is analyzed in this paper, and the stress level of the humans is observed and analyzed for variations based on the biomarkers. The stress levels are monitored and assessed in a controlled and everyday environment. An

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ideal number of 10 features are used for this analysis that is selected by the kNN classifier and a greedy stepwise algorithm. Simulation results show that combining these sensors and features offers improved performance. The system provides reasonable prediction accuracy of 97% and an accuracy of 93% in an everyday environment.

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